CHRISTIAN CARRICO SPICE 005 #2 9/11/2026

$$\left[\begin{array}{ccc|c} 1 & 1 & -1 & 0 \\ 0 & 1 & 0 & E_0/R_1 \\ 0 & 1 & R_2/R_1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right] \Rightarrow$$

$$\left[\begin{array}{ccc|c} 1 & 1 & -1 & 0 \\ 0 & 1 & 0 & E_0/R_1 \\ 0 & 0 & R_2/R_1 & -E_0/R_1 \\ 0 & 0 & 0 & 0 \end{array}\right] \Rightarrow \quad \frac{-E_0/R_1}{R_2/R_1} = \frac{-E_0 R_1}{R_1 R_2} = \frac{-E_0}{R_2}$$

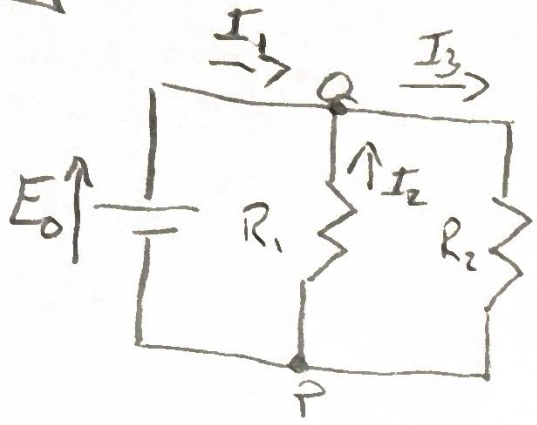
$$\left[\begin{array}{ccc|c} 1 & 1 & -1 & 0 \\ 0 & 1 & 0 & E_0/R_1 \\ 0 & 0 & 1 & -E_0/R_2 \\ 0 & 0 & 0 & 0 \end{array}\right] \Rightarrow \quad -E_0/R_2 - E_0/R_1$$

$$-E_0\left(1/R_2 + 1/R_1\right)$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & -E_0/R_2 - E_0/R_1 \\ 0 & 1 & 0 & E_0/R_1 \\ 0 & 0 & 1 & -E_0/R_2 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$$\boxed{I_1 = -E_0/R_2 - E_0/R_1 \, A \quad I_2 = E_0/R_1 \, A \quad I_3 = E_0/R_3 \, A}$$

19)



$\Rightarrow$

$Q: I_2 + I_1 - I_3 = 0$

$P: I_3 - I_1 - I_2 = 0 \Rightarrow$

$R_1 I_2 + R_2 I_3 = 0$

$R_1 I_2 = E_0$

$$\left[\begin{array}{ccc|c} 1 & 1 & -1 & 0 \\ -1 & -1 & 1 & 0 \\ 0 & R_1 & R_2 & 0 \\ 0 & R_1 & 0 & E_0 \end{array}\right] \Rightarrow$$

$$\left[\begin{array}{ccc|c} 1 & 1 & -1 & 0 \\ 0 & R_1 & R_2 & 0 \\ 0 & R_1 & 0 & E_0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$$\left[\begin{array}{ccc|c} 1 & 1 & -1 & 0 \\ 0 & 1 & 0 & E_0/R_1 \\ 0 & R_1 & R_2 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right] \Rightarrow$$

6|

$$\vec{A} = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix} \Rightarrow$$

$$\begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1-\lambda & 2 \\ 0 & 3-\lambda \end{bmatrix} \Rightarrow$$

$$(1-\lambda)(3-\lambda) - 2(0) \Rightarrow$$

$$\lambda^2 - 4\lambda + 3 = 0$$

$$(\lambda - 3)(\lambda - 1) = 0$$

$$\boxed{\lambda = 3, 1}$$

$$\vec{A}\vec{x} = \lambda\vec{x}$$

$\lambda = 3$

$$\begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 3 \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \Rightarrow$$

$x_1 + 2x_2 = 3x_1 \Rightarrow 2x_2 = 2x_1$ Set $x_2 = C$, $\Rightarrow$ $x_1 = C$

$3x_2 = 3x_2 \quad x_2 = x_2$ then $x_1 = C$ $x_2 = C$

$$\boxed{\lambda = 3, \ \vec{x}_1 = \begin{bmatrix} C \\ C \end{bmatrix}}$$

$\lambda = 1$

$$\begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 1 \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \Rightarrow$$

$x_1 + 2x_2 = x_1$

$3x_2 = x_2$ $\Rightarrow x_2 = 0 \Rightarrow$ Set $x_1 = C \Rightarrow$ $\boxed{\lambda = 1, \ \vec{x}_2 = \begin{bmatrix} C \\ 0 \end{bmatrix}}$

3)

$$A = \begin{bmatrix} 7 & \sqrt{6} \\ \sqrt{6} & 2 \end{bmatrix} \qquad A - \lambda I = \begin{bmatrix} 7-\lambda & \sqrt{6} \\ \sqrt{6} & 2-\lambda \end{bmatrix}$$

$$\det(A - \lambda I) = (7-\lambda)(2-\lambda) - \sqrt{6}^2 =$$
$$14 - 9\lambda + \lambda^2 - \sqrt{6}^2 =$$
$$8 - 9\lambda + \lambda^2 =$$
$$(\lambda - 1)(\lambda - 8) = 0$$
$$\boxed{\lambda = 1, 8}$$

$\lambda = 8$

$$\begin{bmatrix} 7 & \sqrt{6} \\ \sqrt{6} & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 8 \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$7x_1 + \sqrt{6}x_2 = 8x_1$ $\Rightarrow$ $\sqrt{6}x_2 = x_1$ let $x_2 = c$

$\sqrt{6}x_1 + 2x_2 = 8x_2$

$$\boxed{\vec{x}_1 = \begin{bmatrix} \sqrt{6}c \\ c \end{bmatrix}}$$

$\lambda = 1$

$$\begin{bmatrix} 7 & \sqrt{6} \\ \sqrt{6} & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$7x_1 + \sqrt{6}x_2 = x_1$ $\rightarrow$ $-\sqrt{6}x_1 = x_2$ let $x_1 = c$

$\sqrt{6}x_1 + 2x_2 = x_2$

$$\boxed{\vec{x}_2 = \begin{bmatrix} c \\ -\sqrt{6}c \end{bmatrix}}$$

3) (Direction)

$$\vec{X}_1 = \begin{bmatrix} \sqrt{6}c \\ c \end{bmatrix} \qquad \vec{X}_2 = \begin{bmatrix} c \\ -\sqrt{6}c \end{bmatrix}$$

$\vec{X}_1$ is in QI

$$\tan(\phi) = \left(\frac{c}{\sqrt{6}c}\right) = \left(\frac{1}{\sqrt{6}}\right) \Rightarrow \phi = \tan^{-1}\left(\frac{1}{\sqrt{6}}\right) = \boxed{22.21^\circ}$$

$\vec{X}_2$ is in QIV

$$\tan(\phi) = \left(\frac{-\sqrt{6}c}{c}\right) = (-\sqrt{6}) \Rightarrow \phi = \tan(-\sqrt{6}) = \boxed{-67.79^\circ \text{ or } 112.2^\circ}$$

1)

$$\vec{A} = \begin{bmatrix} 0.8 & 0.6 \\ -0.6 & 0.8 \end{bmatrix}$$

$$\vec{A}^T = \begin{bmatrix} 0.8 & -0.6 \\ 0.6 & 0.8 \end{bmatrix} \qquad -\vec{A} = \begin{bmatrix} -0.8 & -0.6 \\ 0.6 & -0.8 \end{bmatrix}$$

$$\vec{A}^{-1} = \frac{1}{0.8^2 - (-0.6)(0.6)} \begin{bmatrix} 0.8 & -0.6 \\ 0.6 & 0.8 \end{bmatrix} = \begin{bmatrix} 0.8 & -0.6 \\ 0.6 & 0.8 \end{bmatrix}$$

$\vec{A} \neq \vec{A}^T$, $\boxed{\vec{A} \text{ is not symmetric}}$

$\vec{A}^T \neq -\vec{A}$, $\boxed{\vec{A} \text{ is not skew symmetric}}$

$\vec{A}^T \neq \vec{A}^{-1}$, $\boxed{\vec{A} \text{ is orthagonal}}$

$$\vec{A} = \begin{bmatrix} 0.8 & 0.6 \\ -0.6 & 0.8 \end{bmatrix}$$

$$Det(\vec{A} - \lambda I) = (0.8-\lambda)^2 - (0.6)(-0.6) = \lambda^2 - 1.6\lambda + 0.64 + 0.36 =$$

$$\lambda^2 - 1.6\lambda + 1 = 0$$

$$\boxed{\lambda = 0.8 \pm 0.6i}$$

For $\lambda = 0.8 + 0.6i$

$$\begin{bmatrix} 0.8 & 0.6 \\ -0.6 & 0.8 \end{bmatrix} \begin{bmatrix} X_1 \\ X_2 \end{bmatrix} = 0.8+0.6i \begin{bmatrix} X_1 \\ X_2 \end{bmatrix}$$

$$0.8x_1 + 0.6x_2 = (0.8+0.6i)x_1$$
$$-0.6x_1 + 0.8x_2 = (0.8+0.6i)x_2 \Rightarrow -0.6x_1 = 0.6ix_2$$

let $x_1 = C$

$$\boxed{\vec{X}_1 = \begin{bmatrix} C \\ iC \end{bmatrix}}$$

For $\lambda = 0.8 - 0.6i$

$$0.8x_1 + 0.6x_2 = (0.8-0.6i)x_1$$
$$-0.6x_1 + 0.8x_2 = (0.8-0.6i)x_2 \Rightarrow -0.6x_1 = 0.6x_2$$

let $x_1 = C$

$$\boxed{\vec{X}_2 = \begin{bmatrix} C \\ -iC \end{bmatrix}}$$